

SANTIAGO DE COMPOSTELA AP, SpainWMO: **080420**Lat: **42.888N**Lon: **8.411W**Elev: **1213**StdP: **14.06**Time Zone: **1.00 (EUC)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	31.6	33.8	25.0	19.9	39.2	28.2	23.2	39.6	25.0	52.2	22.5	51.5	3.3	100	0.495

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	19.0	86.9	68.1	82.7	66.6	79.0	65.2	70.2	82.8	68.3	79.5	66.6	76.4	6.5	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.2	100.8	74.4	64.5	95.2	71.7	63.0	90.1	69.3	34.9	82.9	33.2	79.7	31.8	75.5	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.0	17.3	15.3	DB	26.5	94.8	2.4	3.8	24.8	97.5	23.4	99.8	22.0	101.9	20.3	104.7	(4)
(5)			WB	25.9	74.6	2.3	2.8	24.3	76.6	22.9	78.2	21.7	79.8	20.0	81.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.6	46.2	46.8	50.5	52.6	56.9	62.4	65.5	66.2	63.2	58.0	50.4	47.5	(6)	
	DBStd	8.85	4.68	4.92	5.50	5.90	5.72	5.58	5.08	4.63	4.94	5.19	5.07	4.87	(7)	
	HDD50	514	133	111	62	37	6	0	0	0	0	4	54	107	(8)	
	HDD65	3687	582	511	450	374	262	114	54	40	92	225	439	544	(9)	
	CDD50	2541	16	20	76	114	218	373	482	501	396	251	66	28	(10)	
	CDD65	240	0	0	0	2	9	37	71	76	38	7	0	0	(11)	
	CDH74	2288	0	0	8	49	152	421	625	626	341	66	0	0	(12)	
	CDH80	691	0	0	0	4	33	138	213	200	93	10	0	0	(13)	
(14) Wind	WSAvg	6.7	7.0	6.9	7.3	7.2	6.8	6.6	6.4	6.3	5.7	6.1	6.5	7.0	(14)	
(15) Precipitation	PrecAvg	62.0	8.3	5.4	5.3	5.2	4.4	2.1	1.5	2.0	3.5	8.6	8.0	7.7	(15)	
	PrecMax	82.9	17.8	14.8	22.9	15.2	10.4	5.4	5.1	7.4	10.3	19.0	16.7	23.4	(16)	
	PrecMin	40.6	1.3	0.0	0.0	1.6	1.2	0.0	0.0	0.0	0.6	0.0	0.4	0.7	(17)	
	PrecStd	12.5	4.3	3.7	5.0	3.5	2.4	1.7	1.2	1.9	2.5	4.4	4.6	5.0	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.4	67.1	75.0	79.2	84.5	91.1	92.3	91.4	88.2	81.0	67.8	61.2	(19)	
		MCWB	51.9	53.6	56.2	59.9	64.4	69.0	69.0	70.6	65.9	63.6	58.4	53.1	(20)	
	2%	DB	56.9	62.1	70.1	73.9	78.3	83.9	85.8	85.7	82.1	74.6	62.9	58.0	(21)	
		MCWB	52.8	51.5	54.3	58.2	62.2	66.9	69.2	68.3	65.7	61.9	57.2	54.0	(22)	
	5%	DB	55.1	57.3	65.1	68.5	73.3	78.7	80.6	80.7	77.4	69.9	60.5	56.6	(23)	
		MCWB	52.4	50.0	52.7	56.3	60.7	65.0	67.8	66.9	64.1	60.3	56.7	53.7	(24)	
	10%	DB	53.4	54.0	60.0	63.0	67.9	73.7	76.5	76.7	73.3	66.2	57.5	55.0	(25)	
		MCWB	51.3	50.3	51.5	53.8	58.5	63.1	65.5	65.4	62.7	59.8	54.8	52.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.4	56.2	58.5	62.1	67.0	71.6	72.5	73.3	69.6	66.7	61.4	57.6	(27)	
		MCDB	57.1	61.3	66.4	75.0	79.6	83.9	86.6	86.9	82.3	73.1	63.3	58.1	(28)	
	2%	WB	54.9	54.2	56.2	59.2	63.8	68.2	70.1	70.1	67.2	64.4	59.2	56.1	(29)	
		MCDB	55.6	57.3	64.4	71.4	75.2	80.9	82.8	81.8	78.1	69.7	61.7	56.9	(30)	
	5%	WB	53.3	52.8	54.5	57.1	61.5	65.8	68.0	67.9	65.3	62.7	57.2	54.6	(31)	
		MCDB	54.3	55.7	61.8	66.9	70.9	76.9	79.2	77.8	74.4	66.9	59.2	55.7	(32)	
	10%	WB	51.7	51.2	53.0	55.1	59.2	63.6	66.0	66.0	63.8	61.0	55.5	52.9	(33)	
		MCDB	52.9	54.0	58.8	61.7	66.0	72.2	75.0	74.1	71.2	64.9	57.3	54.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	11.4	13.9	16.3	16.9	17.4	18.5	18.8	19.0	18.2	14.9	12.0	11.5	(35)	
		MCDBR	13.5	21.3	26.7	27.9	27.8	28.4	27.6	27.3	26.2	22.6	15.9	13.6	(36)	
	5% WB	MCWBR	10.0	12.8	14.2	14.5	14.6	14.3	13.3	12.7	12.3	11.9	10.5	10.1	(37)	
		MCWBR	9.3	14.6	20.3	23.7	24.4	25.1	24.8	23.5	22.5	16.5	11.7	9.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.324	0.330	0.362	0.366	0.371	0.374	0.363	0.365	0.357	0.351	0.330	0.317	(40)	
		taud	2.489	2.456	2.372	2.366	2.375	2.383	2.417	2.419	2.445	2.474	2.490	2.501	(41)	
	Ebn at Noon		258	276	279	286	286	284	286	283	277	264	253	250	(42)	
		Edh at Noon	23	28	34	37	38	37	36	35	31	27	23	21	(43)	
(44) All-Sky Solar Radiation	RadAvg		476	773	1129	1522	1832	2029	2039	1839	1450	907	556	435	(44)	
	RadStd		71	120	179	166	176	141	125	137	94	119	90	64	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+61

Nomenclature: See separate page